Prevention of type 2 diabetes mellitus by changes in lifestyle among subjects with impaired glucose tolerance.

BACKGROUND: Type 2 diabetes mellitus is increasingly common, primarily because of increases in the prevalence of a sedentary lifestyle and obesity. Whether type 2 diabetes can be prevented by interventions that affect the lifestyles of subjects at high risk for the disease is not known. METHODS: We randomly assigned 522 middle-aged, overweight subjects (172 men and 350 women; mean age, 55 years; mean body-mass index [weight in kilograms divided by the square of the height in meters], 31) with impaired glucose tolerance to either the intervention group or the control group. Each subject in the intervention group received individualized counseling aimed at reducing weight, total intake of fat, and intake of saturated fat and increasing intake of fiber and physical activity. An oral glucose-tolerance test was performed annually; the diagnosis of diabetes was confirmed by a second test. The mean duration of follow-up was 3.2 years. RESULTS: The mean (+/-SD) amount of weight lost between base line and the end of year 1 was 4.2+/-5.1 kg in the intervention group and 0.8+/-3.7 kg in the control group; the net loss by the end of year 2 was 3.5+/-5.5 kg in the intervention group and 0.8+/-4.4 kg in the control group (P<0.001 for both comparisons between the groups). The cumulative incidence of diabetes after four years was 11 percent (95 percent confidence interval, 6 to 15 percent) in the intervention group and 23 percent (95 percent confidence interval, 17 to 29 percent) in the control group. During the trial, the risk of diabetes was reduced by 58 percent (P<0.001) in the intervention group. The reduction in the incidence of diabetes was directly associated with changes in lifestyle. CONCLUSIONS: Type 2 diabetes can be prevented by changes in the lifestyles of high-risk subjects.